School of Information Science

Course Plan - Lab/Practice-based Course

Academic Year 2025-26 EVEN SEMESTER

School/Department of Students	School of Information Science
Name of the Program(s) of Students	BCA(Bachelor of Computer Applications), BCA(Bachelor of Computer Applications)(Artificial Intelligence and Machine Learning), BCA(Bachelor of Computer Applications)(Data Science)
PRC Approval Ref. No.	
Semester/Year	IV/II
Course Code & Name	CSA3423 & NoSQL
Credit Structure (L-T-P-C)	1-0-4-3
Contact Hours	75 Sessions
Course In-Charge (IC)	Krishna Mishra
Course Instructor(s)	Dr. Salma Firdose, Mr. Akshay GS, Mr. Jatin, Mr. Suhel Khan
Course URL	https://presidencyuniversity.linways.com

1. COURSE PRE-REQUISITES:
Basic understanding of database concepts. Familiarity with SQL and relational database management systems.

2. COURSE DESCRIPTION:
This course provides an in-depth understanding of NoSQL databases, their architecture, and their applications in modern data-driven environments. Students will explore the key concepts, types, and use cases of NoSQL databases, focusing on their scalability, flexibility, and performance advantages over traditional relational databases. The course covers various NoSQL database models, including Key-Value, Document-Oriented, Column-Family, and Graph Databases, with practical examples and hands-on experience. Students will gain the skills to design, implement, and manage NoSQL databases for real-world applications such as Big Data, IoT, and E-commerce systems.

3. COURSE OBJECTIVES:

The course No SQL aims to equip BCA students with foundational knowledge and practical skills in NoSQL databases, focusing on their architecture, types, and applications. Students will learn to design, implement, and manage scalable, distributed systems.

4. COURSE OUTCOMES:

TABLE 1: COURSE OUTCOMES		
CO Number	**Statement of CO** *On successful completion of the course the students shall be able to*	**Blooms Cognitive Level**
CO1	Understand NoSQL Fundamentals	Understand
CO2	Perform Practical NoSQL Operations	Apply
CO3	Design Scalable Systems	Create
CO4	Apply NoSQL in Real-World Scenarios	Apply

5. MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

5.1 *PROGRAM OUTCOMES*:

On successful completion of the Program, the students will be able to:

PO1. Application of Domain Knowledge: Apply the domain knowledge such as mathematics, science and software engineering fundamentals into the Computer Application related professions.

PO2. Problem Solving & Analysis: Identify, Formulate, Analyze and Solve Complex Scenarios related to Computer Applications.

PO3. Design/development of Activities: Conceive, Design and Develop various activities of Computer Applications.

PO4. Conduct Investigations of Events: Carry out Investigation of an event and draw logical conclusions based on critical thinking and analytical reasoning

PO5. Modern Tool usage: Effectively apply relevant ICT Tools and digital tools to carry out Computer Application Attributes.

PO6. Research: Identify suitable Research Methods and report the findings.

PO7. Profession and Society: Apply the knowledge of the values and beliefs of multicultural society and a global perspective in the profession.

PO8. Ethics: Identify ethical issues and embrace ethical values in conduct of Profession.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Express thoughts and ideas effectively in writing and oral communication

PO11. Project management and finance: Ability to work independently, identify appropriate resources required for a project, and manage a project through to completion.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of societal and technological change

TABLE 2a: CO-PO Mapping

CO. No	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	L	L	-	-	-	-	-	-	-	-	-	-
CO2	H	M	L	-	-	-	-	-	-	-	-	-
CO3	H	M	L	L	H	-	-	-	-	-	-	M
CO4	H	H	M	-	-	-	-	-	M	M	-	M

5.2 PROGRAM SPECIFIC OUTCOMES:

On successful completion of the Program, the students will be able to:

PSO1	[Disciplinary knowledge]: Capable of demonstrating comprehensive knowledge and understanding of Computer Applications, Data Science and AI/ML techniques.
PSO2	[Problem Solving]: Identify, formulate and apply appropriate techniques in the areas related to Software development, big data, Network, Cloud computing technologies and related domains of varying complexities in real-time applications.
PSO3	[Design/development of Applications]: Conceive, Design and Develop Various Computer Applications and be able to apply data science and AI/ML techniques in specific domains such as healthcare, finance, agriculture, marketing, etc.,

TABLE 2b: CO-PSO Mapping

CO Number	PSO1	PSO2	PSO3
CO1	M	L	-
CO2	H	M	L
CO3	H	M	L
CO4	H	M	L

6. DETAILS OF THE EXPERIMENTS/PRACTICAL SESSIONS:

a. **Theory:**

Module Number	Module Name	Number of Sessions
1.	Introduction to NoSQL Databases Overview of NoSQL and its importance, Differences between SQL and NoSQL databases, Installation and setup of NoSQL databases (MongoDB, Cassandra, Redis, Neo4j), Introduction to basic NoSQL commands.	3
2.	Document-Oriented and Key-Value Databases Understanding document-oriented databases (MongoDB), CRUD operations in MongoDB, Data modeling and schema design in MongoDB, Introduction to key-value stores (Redis), Working with Redis data structures (strings, lists, sets, and hashes).	4
3.	Column-Family Databases (Cassandra) Introduction to column-family databases and their architecture, Basics of Apache Cassandra and its use cases, creating keyspaces and tables in Cassandra, Performing CRUD operations using CQL (Cassandra Query Language), Data partitioning, replication, and consistency in Cassandra.	4
4.	Graph Databases (Neo4j) Introduction to graph databases and their applications Nodes, relationships, and properties in Neo4j , Querying graph databases using Cypher ,Real-world use cases of graph databases, Indexing, aggregation, and performance optimization in MongoDB, Sharding and replication in NoSQL databases, Security and access control in NoSQL databases	4

***6.b* Practical/ Laboratory:**

S. No	Name of the Experiment/Exercise	Number of Sessions
1.	Install MongoDB locally and connect using Mongo Shell.	2
2.	Install Cassandra and verify cluster setup.	2
3.	Install Redis and perform basic commands (SET, GET).	2
4.	Install Neo4j Desktop and explore Neo4j Browser.	2
5.	Compare SQL vs NoSQL data models using simple examples	2

6.	Execute basic commands (create database, insert record, retrieve) in all four NoSQL databases.	2
7.	Create a MongoDB collection and insert multiple documents.	2
8.	Perform CRUD operations on MongoDB documents.	2
9.	Design a MongoDB schema for an e-commerce application (products, users, orders).	2
10.	Perform indexing in MongoDB for faster search.	2
11.	Insert and retrieve different data types (strings, lists, sets, hashes) in Redis.	2
12.	Implement expiration (TTL) of keys in Redis.	2
13.	Simulate a simple leaderboard using Redis Sorted Sets.	2
14.	Create a session store system using Redis (e.g., for login sessions).	2
15.	Create a keyspace and table in Cassandra using CQL.	2
16.	Insert and retrieve data using Cassandra Query Language (CQL).	2
17.	Implement a Student Management System database in Cassandra.	2
18.	Demonstrate partitioning by inserting data with different partition keys.	2
19.	Demonstrate replication by setting replication factor in Cassandra.	2
20.	Query Cassandra using SELECT, WHERE clauses with clustering columns.	2
21.	Simulate a distributed database setup using multiple Cassandra nodes (pseudo/multinode setup locally).	2
22.	Analyze Consistency Level settings (ONE, QUORUM, ALL) in Cassandra queries.	2
23.	Create nodes and relationships in Neo4j (e.g., Students-Friends, Company-Employee).	2
24.	Perform basic Cypher queries: MATCH, CREATE, RETURN.	2
25.	Design a Movie Recommendation Graph database in Neo4j.	2

26.	Use Neo4j to model a social network (likes, follows, comments).	2
27.	Execute aggregation queries in Neo4j (e.g., count relationships).	2
28.	Perform indexing and constraint creation in Neo4j for optimization.	2
29.	Create complex Cypher queries: optional matches, patterns, subqueries.	2
30.	Analyze real-world case study: Supply Chain Management graph using Neo4j.	2

REFERENCE MATERIALS:

Text Books:

T1: NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence by Pramod J. Sadalage and Martin Fowler.

T2: MongoDB: The Definitive Guide by Kristina Chodorow.

Reference Books:

R1: Cassandra documentation: https://cassandra.apache.org/doc/latest/.

R2: Adam Fowler, "NoSQL For Dummies", Wiley, 2025.

Online Resources

1. https://www.mongodb.com/resources/basics/databases/nosql-explained

Practical/Laboratory Resources:

1. NoSQL Laboratory Manual, Presidency University, Bengaluru

7. SCHEDULE OF INSTRUCTION

7.a Detailed Schedule of Instructions: Theory

TABLE 3a: LESSON PLAN: THEORY				
Session Number	**Topic**	**Sub-Topic**	**CO Number**	**Reference**
1.	Program Integration & Course Integration	Overview of the Course, Scopes and Opportunities of T and D	CO1	T1
Module 1				
2.	Introduction to NoSQL Databases	Overview of NoSQL and its importance, Differences between SQL and NoSQL databases.	CO1	T1: Ch. No. 1 Pg:1-4 Ch. No. 2 Pg:15-28
3.	Introduction to NoSQL Databases	Installation and setup of NoSQL databases (MongoDB, Cassandra, Redis, Neo4j).	CO1	Ch. No. 3 Pg:29-44
4.	Introduction to NoSQL Databases	Introduction to basic NoSQL commands.	CO2	T1: Ch. No.3 Pg:29-44 T1: Ch. No.4 Pg:45-60

5.	Mid Term			
Module 2				
5.	Document-Oriented and Key-Value Databases	Understanding document-oriented databases (MongoDB).	CO2	T1: Ch. No. 2 Pg No: 15-28, 29-44
6.	Document-Oriented and Key-Value Databases	CRUD operations in MongoDB, Data modeling and schema design in MongoDB	CO1	T1: Ch. No. 2 Pg No: 15-28, Ch. No. 6 Pg. No. 29-44
7.	Document-Oriented and Key-Value Databases	Introduction to key-value stores (Redis)	CO2	T1: Ch. No. 3 Pg No: 29-44
8.	Document-Oriented and Key-Value Databases	Working with Redis data structures (strings, lists, sets, and hashes).	CO2	T1: Ch. No. 3 Pg No: 29-44, Ch. No. 4 Pg. No. 45-60
Midterm Exam Question Paper and Scheme of Evaluation – Discussion				
Module 3				
8.	Column-Family Databases (Cassandra)	Introduction to column-family databases and their architecture	CO2	T1: Ch. No. 3 Pg. No. 29-44 Ch. No. 4 Pg. No. 45-60
9.	Column-Family Databases (Cassandra)	Basics of Apache Cassandra and its use cases.	CO2	T1: Ch. No. 4 Pg. No. 45-60 Ch. No. 5 Pg. No. 61-72
10.	Column-Family Databases (Cassandra)	Creating keyspaces and tables in Cassandra, Performing CRUD operations using CQL (Cassandra Query Language)	CO4	T1: Ch. No. 3 Pg. No. 29-44
11.	Column-Family Databases (Cassandra)	Data partitioning, replication, and consistency in Cassandra.	CO4	T1: Ch. No. 5 Pg. No. 61-72 Ch. No. 6 Pg. No. 73-88
Module 4				
12.	Graph Databases (Neo4j)	Introduction to graph databases and their applications Nodes, relationships, and properties in Neo4j, Querying graph databases using Cypher	CO4	T1: Ch. No. 3 Pg. No. 29-44

13.	Graph Databases (Neo4j)	Real-world use cases of graph databases, Indexing, aggregation, and performance optimization in MongoDB,	CO3	T1: Ch. No. 7 Pg. No. 89-104
14.	Graph Databases (Neo4j)	Sharding and replication in NoSQL databases,	CO4	T1: Ch. No. 5 Pg. No. 61-72 Ch. No. 4 Pg. No. 45-60
15.	Graph Databases (Neo4j)	Security and access control in NoSQL databases	CO4	T1: Ch. No. 6 Pg. No. 73-88 Ch. No. 7 Pg. No. 89-104

7.b Detailed Schedule of Instructions: Practical

TABLE 3b: LESSON PLAN: PRACTICAL		
Session No	**Experiment Title**	**Mapped CO**
P1	Install MongoDB locally and connect using Mongo Shell.	CO1
P2	Install Cassandra and verify cluster setup.	CO1
P2	Install Redis and perform basic commands (SET, GET).	CO1
P3	Install Neo4j Desktop and explore Neo4j Browser.	CO1
P5	Compare SQL vs NoSQL data models using simple examples	CO1
P6	Execute basic commands (create database, insert record, retrieve) in all four NoSQL databases.	CO1
P7	Create a MongoDB collection and insert multiple documents.	CO1
P8	Perform CRUD operations on MongoDB documents.	CO1
P9	Design a MongoDB schema for an e-commerce application (products, users, orders).	CO3
P10	Perform indexing in MongoDB for faster search.	CO3
P11	Insert and retrieve different data types (strings, lists, sets, hashes) in Redis.	CO2
P12	Implement expiration (TTL) of keys in Redis.	CO2
P13	Simulate a simple leaderboard using Redis Sorted Sets.	CO4
P14	Create a session store system using Redis (e.g., for login sessions).	CO4
P15	Create a keyspace and table in Cassandra using CQL.	CO1
P16	Insert and retrieve data using Cassandra Query Language (CQL).	CO1
P17	Implement a Student Management System database in Cassandra.	CO4

P18	Demonstrate partitioning by inserting data with different partition keys.	CO3
P19	Demonstrate replication by setting replication factor in Cassandra.	CO3
P20	Query Cassandra using SELECT, WHERE clauses with clustering columns.	CO2
P21	Simulate a distributed database setup using multiple Cassandra nodes (pseudo/multinode setup locally).	CO3
P22	Analyze Consistency Level settings (ONE, QUORUM, ALL) in Cassandra queries.	CO3
P23	Create nodes and relationships in Neo4j (e.g., Students-Friends, Company-Employee).	CO2
P24	Perform basic Cypher queries: MATCH, CREATE, RETURN.	CO2
P25	Design a Movie Recommendation Graph database in Neo4j.	CO4
P26	Use Neo4j to model a social network (likes, follows, comments).	CO4
P27	Execute aggregation queries in Neo4j (e.g., count relationships).	CO2
P28	Perform indexing and constraint creation in Neo4j for optimization.	CO3
P29	Create complex Cypher queries: optional matches, patterns, subqueries.	CO3
P30	Analyze real-world case study: Supply Chain Management graph using Neo4j.	CO4

The main pedagogical methods in the course are as follows:

- Lecture mode.
- Power Point Presentation.
- Demonstration-Based Learning
- Experiential (Hands-On) Learning
- Inquiry-Based Learning (IBL)
- Collaborative/Peer Learning
- Problem-Based Learning (PBL)
- Simulation + Real-Lab Hybrid Learning.

TABLE 4: SPECIAL DELIVERY METHOD			
S. No	**Session Number** ***(Mention Theory/Practical clearly)***	**Subtopic (as per lesson plan)**	**Pedagogical Method**
1.	S13	Real-world use cases of graph databases, Indexing, aggregation, and performance optimization in MongoDB	Experiential (Hands-On) Learning - Students perform a **hands-on lab** to create collections and apply **single-field and compound indexes**.

2.	P9	Design a MongoDB schema for an e-commerce application (products, users, orders).	Problem-Based Learning (PBL) - Students are given a **real-world problem statement**: *Design a scalable e-commerce database*.
3.	P23	Create nodes and relationships in Neo4j (e.g., Students-Friends, Company-Employee).	Problem-Based Learning (PBL) - Students receive a **scenario-based task** (e.g., model a social network or organizational hierarchy).
4.	P30	Analyze real-world case study: Supply Chain Management graph using Neo4j.	Problem-Based Learning (PBL) - Students are provided with a **real-world supply chain case** (suppliers, warehouses, distributors, retailers).

8. ASSESSMENT

TABLE 5: DETAILS OF ASSESSMENT

Sl. No	Assessment type	Content	Course outcome Number	Duration	Marks	Weightage
1.	Continuous Assessment (CA)	Module-1	CO1	60	100	50%
2.	Midterm Exam: Theory & Practical	Module-1&2	CO1	90	50	25%
3.	End Term Examination: Theory & Practical	Module-1,2,3&4	CO1, CO2, CO3, CO4	180	50	25%

*** Please include Rubrics, if any, for evaluating practical component.**

9. COURSE CLEARANCE CRITERIA:

This is in accordance with the Academic Regulations of the University and the Program Regulations and Curriculum of the respective program.

10. SAMPLE QUESTIONS:

TABLE 6: SAMPLE QUESTIONS

Sl. No	Question	Marks	CO Number	Blooms Cognitive Level
1	MongoDB database creation and CRUD operations for student management	5	CO1	L3
2	Develop a MongoDB-based application to store and manage employee records for an organization. Perform queries to retrieve employees based on department and salary conditions, and demonstrate sorting and projection operations. Discuss how document-oriented databases handle large volumes of data efficiently.	5	CO2	L3
3	Create a suitable MongoDB document schema for an online shopping system that includes customers, products, and orders. Insert sample data and explain the design choices made for scalability and performance.	5	CO3	L6
4	Implement a sales data management system using MongoDB and perform aggregation operations to	5	CO2	L4

	compute total sales, average sales, and identify top-selling products. Explain the role of the aggregation framework in data analysis.			
5	Design and implement a key-value based caching system using Redis to store user session information and recent activity logs. Demonstrate the use of different Redis data structures and explain their advantages in real-time applications.	10	CO4	L3
6	Develop a Redis-based notification system where user profiles are stored using hashes and subscription details are maintained using sets. Perform insert and retrieval operations and justify the choice of Redis for this application.	10	CO4	L4
7	Create a Cassandra keyspace and table to manage course information in a university. Insert sample records and demonstrate CRUD operations using Cassandra Query Language (CQL). Explain the importance of replication and consistency in Cassandra.	10	CO2	L3
8	Design a Cassandra database schema for storing time-series sensor data. Demonstrate the use of partition keys and clustering columns, and explain how data partitioning improves performance and scalability.	10	CO3	L6
9	Implement a graph database using Neo4j for a social networking application. Create nodes and relationships, and write Cypher queries to retrieve friends of a user and identify mutual connections. Explain how graph databases efficiently handle relationship-based queries.	10	CO4	L4
10	Design a Neo4j graph model for a transportation or logistics network. Write Cypher queries to find connected nodes and shortest paths, and explain why graph databases are suitable for real-world network-based applications.	20	CO3	L5

11. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

TABLE 7: SDG MAPPING

S. No	Topic	SDG Number	Justification
1	Introduction to NoSQL Databases	SDG 4	Provides quality education by exposing students to modern and scalable database technologies.
2	Column-Family Databases (Cassandra)	SDG 9	Supports innovation and robust digital infrastructure through scalable and fault-tolerant databases.
3	Data Modeling and Storage Optimization in NoSQL	SDG 12	Promotes responsible data management through efficient storage and optimized resource usage.

12. CRITERIA FOR COURSE OUTCOME ATTAINMENT CALCULATION:

TABLE 8: Threshold and Target Set for Course Outcomes

Sl. No	C.O. No.	Course Outcomes	Threshold in %	Target in %
1.	CO1	Understand NoSQL Fundamentals	65%	70%
3.	CO2	Perform Practical NoSQL Operations	65%	70%

4.	CO3	Design Scalable Systems	60%	65%
5.	CO4	Apply NoSQL in Real-World Scenarios	60%	65%

13. **SUMMARY:**

TABLE 9: SUMMARY OF SCHEDULE				
Sl. No.	**Activity**	**Start date**	**End date**	**Total number of Sessions**
1.	Program Integration & Overview of the course	07.01.2026	07.01.2026	01
2.	Module: 01: Introduction to NoSQL Databases	08.01.2026	08.02.2026	22
3.	Continuous Assessment: 01	09.02.2026	14.02.2026	01
4.	Module: 02: Document-Oriented and Key-Value Databases	16.02.2026	09.03.2026	15
2.	Mid-Term	10.03.2026	14.03.2026	01
3.	Module: 03: Column-Family Databases (Cassandra)	16.03.2026	26.03.2026	10
3.	Continuous Assessment: 02	27.03.2026	01.04.2026	01
4.	Module: 04: Graph Databases (Neo4j)	02.04.2026	06.05.2026	24
4.	End Term	07.05.2026	30.05.2026	-

14. **CONTACT TIMINGS IN THE CHAMBER FOR DISCUSSION**

Students can meet the respective course instructor during the Chamber Consultation Hour to clarify doubts related to the course.

15. **SPECIFIC GUIDELINES TO STUDENTS, IF ANY:**

- Attend all classes regularly.
- Bring a scientific calculator to every class.
- Refer to online study materials and watch the suggested videos available on the NPTEL website.

Name and Signature of the course In-Charge
Mr. Krishna Mishra
Assistant Professor G1/SOIS

APPROVAL:
This course has been duly verified and approved by the Departmental Academic Committee (DAC).

Name and Signature of the Chairperson - DAC